Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/01/2007
PI Name:	Plaut, Karen Ph.D.		
Project Title:	Is Gravity A Continuum? Investigations Of Mechanisms Mediating Gravity-Induced Change In Mammary Metabolism And Maternal Behavior In Pregnant And Lactating Rats.		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	kplaut@msu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	517-355-8383
Organization Name:	Michigan State University		
PI Address 1:	Animal Science Department		
PI Address 2:	1290G Anthony Hall		
PI Web Page:			
City:	East Lansing	**State:**	MI
Zip Code:	48824-1225	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2005	**End Date:**	07/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle		
Flight Assignment:	STS-70 NOTE: Received NCE to 7/31/07, per A. Chu (7/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wade, Charles (United States Army Institute for Surgical Research) Ronca, April Ph.D. (Wake Forest University School of Medicine)		
Grant/Contract No.:	NNA05CP91A ; NCC2-1373		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Homeostasis, the dynamic state of physiological balance, is fundamentally changed in mammals exposed to either microgravity (spaceflight) or hypergravity (centrifugation). In pregnant and lactating mammals, failure to maintain homeostasis exerts pronounced effects on both mothers and their offspring. Our previous data, derived from pregnant rats exposed to either micro- or hypergravity, support: 1) decreased mammary metabolic activity with increased gravitational force; 2) changes in maternal and neonatal energy balance; 3) altered maternal care; and 4) neonatal mortality. Further studies were designed to determine if: 1) the lactation defect was caused by changes in energy balance; 2) the lactation defect could be corrected by injections of prolactin or bromocryptine; or 3) whether global gene expression profiles would allow us to develop an alternate mechanism that explains the lactation defect that occurs in hypergravity. Studies supported by this grant demonstrated that Perturbations in PRL concentration were accompanied by a decrease in PRL receptor (PRLR) mRNA abundance. It is known that lactogenic hormones, PRL and glucocorticoids, induce PRLR. However, our recent study showed that PRL was not able to reverse the HG-induced PRLR suppression. Further experiments were conducted to evaluate whether glucocorticoid supplementation could increase the PRLR in animals exposed to HG. Secondly, preliminary investigations utilizing across tissue gene enrichment analysis identified derangement in a unique set of genes pivotal to lipid synthesis in HG rats, particularly, in the lactating mammary gland. We, therefore, extended our investigations to map-out other alterations in transcriptional crosstalk across key metabolic tissues that are induced by exposure to HG during the transition from pregnancy to lactation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The established novel data set encompassing gene expression signatures and in-vitro incorporation of labeled glucose into lipids across key metabolic tissues is to be utilized to develop a testable and mechanism-based model of the biological system in collaboration with computer engineers. The engineers will assess whether the software designed for embedded systems is effective for interrogation of a complex biological system and test its accuracy in terms of correct predictions or simulations of biological behavior. If successfully developed, this will provide an avenue to explore function of other biological systems and disease states.
Task Progress:	The manuscript on effects of hypergravity on nutrient uptake and neonatal survival was submitted and accepted (See bibliography). The glucocorticoid supplementation study is completed and paper submitted for publication. In consultation with a Bioinformatician, alterations in transcriptional crosstalk across key metabolic tissues that were induced by exposure to HG during the transition from pregnancy to lactation have been identified and the data is being assimilated for publication.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	VanKlompenberg MK, Patel OV, Plaut K. "Hypergravity exposure affects lipid synthesis in rat mammary gland during pregnancy and lactation." American Dairy Science Association and American Society of Animal Science Joint Midwest Meeting, Des Moines, Iowa, March 19-21, 2007. Journal of Animal Science and Journal of Dairy Science(Midwestern section Supp) 2007 Mar;P106. , Mar-2007
Abstracts for Journals and Proceedings	Patel OV, Ronca A, Wade C, Plaut K. "Evidence of down-regulation of key lipogenic transcripts in major adipogenic tissues of periparturient rats exposed to hypergravity." Experimental Biology Annual Meeting, Washington DC, April 28-May 2, 2007. FASEB Journal 2007 May;21: Abstract 771.3. , May-2007
Abstracts for Journals and Proceedings	Dover H, Patel OV, Ronca A, Wade C, Plaut K. "Global gene expression profiling of key adipogenic tissues in hypergravity-exposed lactating rats." Experimental Biology Annual Meeting, Washington DC, April 28-May 2, 2007. FASEB Journal 2007 May; 21: Abstract 644.1. , May-2007
Abstracts for Journals and Proceedings	VanKlompenberg MK, Patel OV, Plaut K. "Transcript abundance of lipogenic enzymes parallels lipid metabolism in the mammary gland of rats exposed to altered gravity during pregnancy." 23rd Annual Meeting, American Society for Gravitational and Space Biology, October 25-28, 2007, Moffett Field, California. Gravitational and Space Biology 2007 Oct;21(1):13. , Oct-2007
Articles in Peer-reviewed Journals	Lintault LM, Zakrzewska EI, Maple RL, Baer LA, Casey TM, Ronca AE, Wade CE, Plaut K. "In a hypergravity environment neonatal survival is adversely affected by alterations in dam tissue metabolism rather than reduced food intake." J Appl Physiol. 2007 Jun;102(6):2186-93. PMID: 17317876 , Jun-2007
Articles in Peer-reviewed Journals	Patel OV, Maple R, Bear L, Ronca A, Wade C, Plaut K. "Tissue-specific variation in expression of prolactin receptor gene subtypes in hypergravity-exposed rats." Journal of Gravitational Physiology 2007;14: 129-30. , Dec-2007
Articles in Peer-reviewed Journals	Zakrzewska E, Maple RL, Lintault L, Casey TM, Wade C, Baer L, Ronca A, Plaut K. "Hypergravity Exposure During Pregnancy Disrupts Homeorhetic Adaptations in Tissue Metabolism Necessary for Successful Lactation in Rat Dams." Journal of Gravitational Physiology (Submitted), August 2007. , Aug-2007
Articles in Peer-reviewed Journals	Patel OV, Zakrzewska E, Maple RL, Baer LA, Ronca AE, Wade CE, Plaut K. "Lipogenesis impaired in periparturient rats exposed to altered gravity is independent of prolactin and glucocorticoid secretion." Eur J Appl Physiol. 2008 Nov;104(5):847-58. Epub 2008 Jul 30. http://dx.doi.org/10.1007/s00421-008-0840-5 ; PubMed PMID: 18665386 (NOTE: originally reported as "American journal of physiology. Regulatory, integrative and comparative physiology, Submitted October 2007.") , Nov-2008
Articles in Peer-reviewed Journals	Casey T, Patel OV, Plaut K. "Transcriptomes reveal alterations in gravity impact circadian clocks and activate mechanotransduction pathways with adaptation through epigenetic change." Physiol Genomics. 2015 Apr;47(4):113-28. http://dx.doi.org/10.1152/physiolgenomics.00117.2014 ; PubMed PMID: 25649141 , Apr-2015
Articles in Peer-reviewed Journals	Alula KM, Resau JH, Patel OV. "Chronic exposure to altered gravity during the pregnancy-to-lactation transition affects abundance of cytoskeletal proteins in the rat mammary gland." Gravit Space Res. 2018 Jul;6(1):58-72. http://gravitationalandspacebiology.org/index.php/journal/article/view/806 , Jul-2018

Articles in Peer-reviewed Journals	Alula KM, Resau JH, Patel OV. "Impact of g-load shift on temporal expression pattern of apoptosis-linked proteins in the rat mammary gland." Gravit Space Res. 2019;7(1):13-21. https://doi.org/10.2478/gsr-2019-0003 , Aug-2019
Articles in Peer-reviewed Journals	Patel OV, Partridge C, Plaut K. "Space environment impacts homeostasis: Exposure to spaceflight alters mammary gland transportome genes." Biomolecules. 2023 May 22;13(5):872. https://doi.org/10.3390/biom13050872 ; PMID: 37238741; PMCID: PMC10216666 , May-2023